Renderoc LA EG

constructive solutions

Cementitious flowable non-shrink micro concrete grout for repair of damaged concrete.

Uses

Renderoc LA EG is designed for the repair of concrete where large-scale reinstatement or casting into congested reinforcement is required. It may also be utilised for thick section grouting of machine bearing plates and other common grouting applications.

Advantages

- High early strength for early removal of formwork and fast return to service.
- High ultimate compressive strengths for dependable bearing support.
- Free flowing and self-compacting requiring no vibration or agitation.
- Non-bleeding and non-segregating requiring no vibration or agitation.
- Suited for placing in nominal thickness of 40 mm up to 150 mm.
- Non-corrosive ingredients protect steel reinforcement.
- Chloride free.
- Good flow characteristics, self-compacting and very good bond to steel & concrete.

Description

Renderoc LA EG is a ready to use cementitious microconcrete, formulated for stability and flowability, its high strengths make it suitable for repair of concrete sections and grouting.

Renderoc LA EG is designed to require the minimum water content to ensure high early strength and long-term durability.

Typical Properties

Properties stated obtained in lab conditions at ~25°C with a water addition of 3.25 litres.

Wet Density	:	~2300 Kg/m^3
Phase	:	Powder
Colour	:	Grey
Initial Setting Time	:	2 – 3.5 Hours
Final Setting Time	:	4 - 5 Hours

Compressive strength

Age (Days)	:	Strength
1	:	> 20 N/mm^2
3	:	> 35 N/mm^2
7	:	> 40 N/mm^2
28	:	> 55 N/mm^2

Instructions for use

Preparation

The unrestrained surface area of the repair must be kept to a minimum. The formwork should be rigid and tight to prevent loss of material and have properly sealed faces to ensure that no water is absorbed from the repair material.

The formwork should include drainage outlets for pre-soaking and, if beneath a soffit, provision for air-venting. Provision must also be made for suitable access points to pour or pump the mixed micro-concrete into place.

Saw cut or cut back the extremities of the repair locations to a depth of at least 10 mm to avoid feather-edging and to Provide a square edge. Break out the complete repair area to a minimum depth of 40 mm up to the sawn edge.

Clean the surface and remove any dust, unsound or contaminated material, plaster, oil, paint, grease, corrosion deposits or algae.

Where breaking out is not required, roughen the surface and remove any laitance by light scrabbling or abrasive blasting.

Oil and grease deposits should be removed by steam cleaning, detergent scrubbing or the use of a proprietary degreaser. The effectiveness of decontamination should then be assessed by a pull-off test.

Expose fully any corroded steel in the repair area and remove all loose scale and corrosion deposits. Steel should be cleaned to a bright condition paying particular attention to the back of exposed steel bars. Abrasive-blasting is recommended for this process.

Where corrosion has occurred due to the presence of chlorides, the steel should be high-pressure washed with clean water immediately after abrasive-blasting to remove corrosion products from pits and imperfections within its surface.

Renderoc LA EG

For additional protection from future corrosion, coat the prepared reinforcing steel with Nitoprime Zincrich.

The cleaned area must be flooded with clean water for at least 2 hours prior to application. Remove all free water immediately before placing Renderoc LA EG.

Mixing

Measure the precise amount of clean water to achieve the desired consistency. (2.9 to 3.25 litres).

Slowly add the powder to the water and mix mechanically with a slow speed drill (max. 500 rpm) or a pan mixer at least 3 min to achieve a homogeneous consistency.

Place the mixed grout within 3 - 5 minutes after mixing to release the air bubbles.

Do not part-mix material. Always fix full bags.

Placing

The mixed repair concrete shall be poured only from one side of the void to eliminate the entrapment of air or surplus pre-soaking water. In vertical repairs, it is preferable to cast from the bottom-up.

Ensure a continuous flow of material at all times until casting has finished.

Curing

Curing of exposed surface should start 30 min after casting using canvas saturated by clean water for 24 hours. Continue curing using Fosroc Chemtech curing compounds for 28 days to obtain the required strengths. In the absence of clean water, treat exposed surfaces with Fosroc curing compounds or any approved curing methods.

Limitations

Never use mechanical vibrators or agitate the material once cast.

Never add any additional water after mixing.

Never add any more than the stated amount of water.

DO not apply at thicknesses below 40mm or above 150mm without first consulting the Fosroc technical department.

High Temperature Working

At Temperatures above 35°C the grout packages should be stored in cool conditions and cold water (<20°C) should be used for mixing the grout. Consult with Fosroc Technical department for further details.

Estimating

Supply and Yield

Renderoc LA EG is supplied in 25 kg plastic bags.

One bag covers approximately 12.5 litres depending on water addition.

Storage

Renderoc LA EG must be stored in a cool, dry store away from sources of heat in the original unopened packaging.

Precautions

Health and safety

For further information consult the Material Safety Data Sheet available for this product.

Fire

Renderoc LA EG is non-flammable.

Important note
Fosroc products are guaranteed against defective materials and manufacture and are sold subject to its standard Conditions for the Supply of Goods and Service. **All Fosroc datasheets are updated on a regular basis. It is the user's responsibility to obtain the recent version.**

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